

Crystal Symmetry — Complete Concept Guide

Symmetry Elements · Point Groups · The Six Crystal Systems · Crystallographic Axes

Reference: Perkins, *Mineralogy*, 3rd Edition — Chapters 9 & 10

Contents

1. What Is Crystal Symmetry?
2. The Four Symmetry Elements (Mirror, Rotation, Inversion, Rotoinversion)
3. How Symmetry Elements Combine — Special Angles & the Six Axis Combinations
4. Point Groups — The 32 Crystal Classes
5. The Six Crystal Systems — Diagnostic Symmetry & Axial Relations
6. Crystallographic Axes & Unit Cell Parameters
7. Flow Chart: How to Identify the Crystal System of Any Crystal
8. Worked Strategy for Exam Questions Like Q.16
9. Quick-Reference Summary Table

1. What Is Crystal Symmetry?

Symmetry describes the relationship between the parts of an object. For crystals, symmetry is not just a visual curiosity — it is a direct reflection of the ordered, repeating internal atomic structure.

"Symmetry is the order in arrangement and orientation of atoms in minerals, and the order in the consequent distribution of mineral properties."

— Zoltai & Stout (1984), quoted in Perkins, *Mineralogy*, p. 239

Because symmetry governs atomic order, it also governs **every physical property** of a crystal — not just its faces, but its hardness, optical behavior, cleavage, and more, in every direction. This is why identifying a crystal's symmetry (from its faces, or from properties like optical behavior) lets mineralogists infer its internal atomic arrangement, and vice versa.

Why this matters for exam questions

Most DMG-style symmetry questions give you a piece of symmetry information (a certain number/type of axes, mirror planes, or an inversion center) and ask you to name the crystal system. Solving them is a matter of matching the given symmetry description against the diagnostic symmetry of each of the six systems — covered in Section 5.

2. The Four Symmetry Elements

Every crystal's external symmetry can be fully described using combinations of just four types of symmetry elements.

(a) Mirror Plane (m)

Mirror plane (m): An imaginary plane that divides a crystal so that one half is the mirror image of the other. The symmetry operation is called *reflection*.

"We call such symmetry reflection, and we call the plane of the imaginary mirror the mirror plane. In shorthand notation, we use the letter m to designate a mirror of symmetry."

— Perkins, *Mineralogy*, p. 239

A face on one side of a mirror has an identical partner at an equal perpendicular distance on the other side, with opposite "handedness." A cube has the maximum number of mirror planes possible for any crystal:

"A perfect sphere has an infinite number of mirror planes, but... crystals cannot have more mirror planes than a cube. Cubes have nine mirrors."

— Perkins, *Mineralogy*, p. 240

(b) Rotation Axis (1, 2, 3, 4, 6)

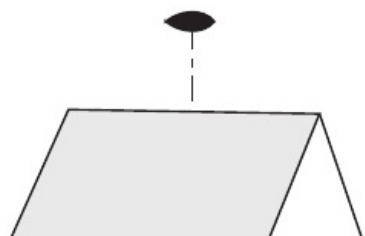
Rotation axis: A line about which a crystal can be rotated and appear unchanged one or more times during a full 360° turn. The symmetry operation is called *rotation*.

"In shorthand notation, we symbolize proper rotation axes using the numbers 1, 2, 3, 4, or 6, corresponding to rotations of 360°, 180°, 120°, 90°, and 60°, respectively."

— Perkins, *Mineralogy*, p. 241

Important restriction

Minerals can only show 1-, 2-, 3-, 4-, or 6-fold rotational symmetry. 5-fold and any symmetry greater than 6-fold (7-fold, 8-fold, etc.) is geometrically impossible for a periodic crystal lattice, even though such shapes can be drawn on paper.



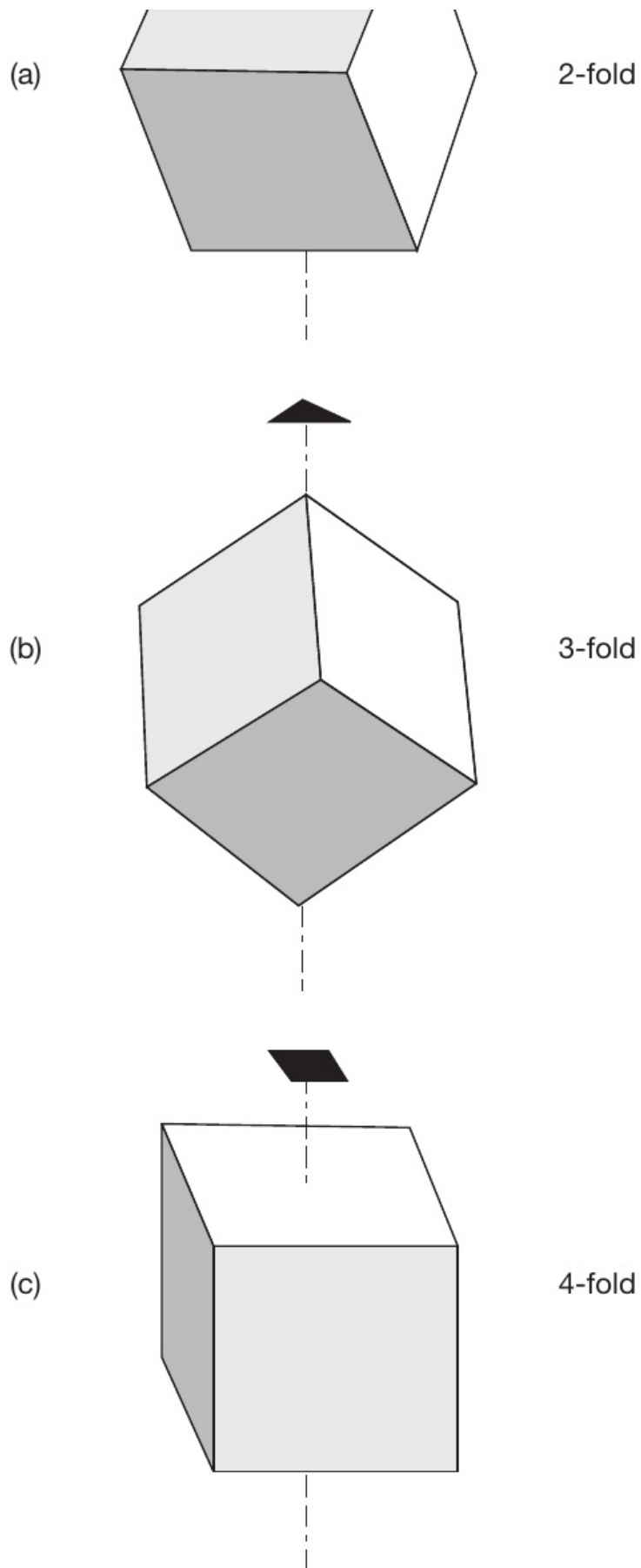


FIGURE 5 Rotation axes of a cube: (a) the edge diagonals of a cube are 2-fold rotational axes of symmetry; (b) the main diagonals of a cube are 3-fold rotational axes of

and 4-fold axes (perpendicular to faces). This is exactly why a cube is used to derive the $54^{\circ}44'$ angle relevant to Q.16.

(c) Inversion Center (i, or $\bar{1}$)

Inversion center: A single point at the center of a crystal through which every face/point is repeated at an equal distance on the opposite side, with reversed orientation (like a pinhole-camera image).

"As with mirror planes, inversion centers relate two identical faces on a crystal; but while mirror planes 'reflect' faces and change their 'handedness,' inversion centers invert them."

— Perkins, *Mineralogy*, p. 241

(d) Rotoinversion Axis ($\bar{1}$, 2, 3, 4, 6)

Rotoinversion: A single, compound symmetry operation that combines rotation and inversion simultaneously — it is **not** simply "rotation, then separately inversion."

"Rotoinversion, a combination of rotation and inversion, is a symmetry operation distinct from the others. The symbols $\bar{1}$, 2, 3, 4 and 6 represent rotoinversion axes. They are articulated as 'bar-1,' 'bar-2,' etc."

— Perkins, *Mineralogy*, p. 246

Key exam fact

Perkins notes: "a 3 operation is equivalent to proper 3-fold rotation and inversion operating separately," and a 2 axis is equivalent to a mirror (m), and a 6 axis is equivalent to 3/m. But **4 is unique** — it cannot be reproduced by any combination of the other basic operations. A crystal can have a 4 axis without having either a true 4-fold axis or an inversion center.

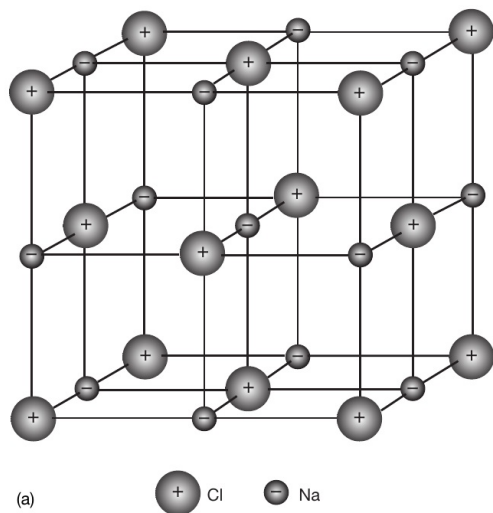


FIGURE 1 (a) The atomic structure of halite showing the cubic arrangement of Na^+ and Cl^- ions. (b) Cubic halite crystals.

the other side of the mirror. The two faces or objects are the same perpendicular distance from the mirror and have opposite “handedness”; if one points to the right, the other points to the left.

Two-dimensional drawings may have many mirror planes or no mirror planes. Circles, for example, contain an infinite number of mirror

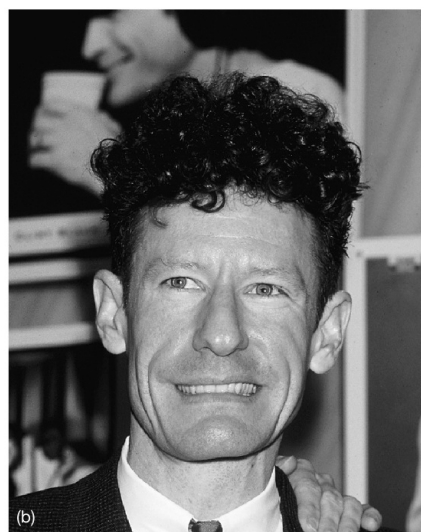
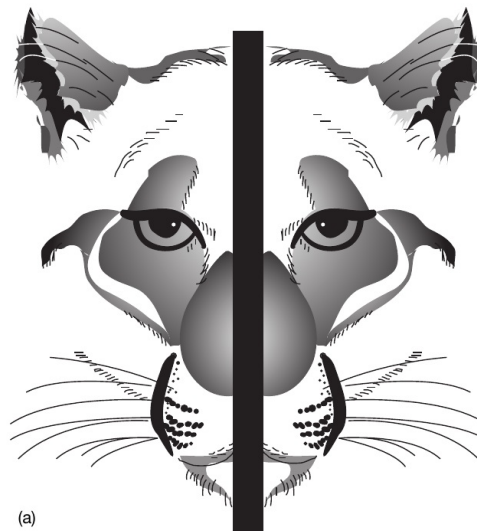


FIGURE 2 Animal, including human, faces are often quite symmetrical. (a) Because the right and left sides of the cougar's face are identical, there is a mirror plane running

Figure 12 & 13 (Perkins, Mineralogy, pp. 246–247) — A 4 rotoinversion axis operating on a point (top), and a side-by-side comparison of a true 4-fold axis, a 4 rotoinversion axis, and a 4-fold axis combined with a separate inversion center ($4/m$) — three genuinely different symmetries that can look superficially similar.

3. How Symmetry Elements Combine — Special Angles

Symmetry elements cannot combine at just any angle. Perkins shows that if two rotation axes intersect at a *non-special* angle (one that doesn't divide evenly into 360°), applying them to each other generates infinitely more axes — which is impossible in a real, finite crystal structure. So rotation axes are only ever found combined at **special angles** (30° , 45° , 60° , 90° , 120° , or the special cube-diagonal angles like $54^\circ 44'$ and $70^\circ 32'$).

This restriction means only **six** combinations of rotation axes are possible in any crystal:

Crystal Morphology and Symmetry

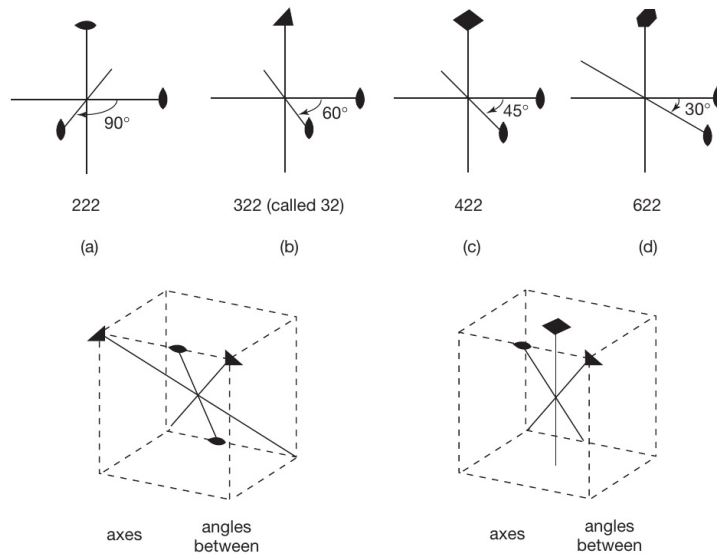


FIGURE 18 The six possible combinations of rotation axes. The angles between axes in (a), (b), (c), and (d) are shown on the drawings. For (e) and (f), based

Figure 18 (Perkins, *Mineralogy*, p. 251) — The six possible combinations of rotation axes: 222, 32, 422, 622 (simple combinations at 90°/60°/45°/30°), and two cube-based combinations (e, f) whose axes are the edges, face diagonals, and body diagonals of a cube — giving the special angles 70°32', 54°44', 45°00', and 35°16'.

$$\text{Angle between a cube edge and a cube body diagonal} = \arccos(1/\sqrt{3}) = 54.7356^\circ = 54^\circ 44'$$

(This is exactly the angle in Q.16, and it is only ever produced by **four** 3-fold axes — the signature of the isometric system.)

4. Point Groups — The 32 Crystal Classes

Point group: The complete set of symmetry operations (rotation axes, mirrors, inversion center, rotoinversion axes) that leave at least one point in the crystal fixed and describe its overall symmetry. There are exactly **32** possible point groups for crystals — no more, no fewer.

Because several symmetry operators are redundant with each other (e.g. a 2 axis behaves exactly like a mirror plane), crystallographers describe all crystal symmetry using just 13 independent standard operators (1, 2, 3, 4, 6, I, 2(=m), 3, 4, 6, i, plus combined symbols like 2/m, 4/m, 6/m). Point group symbols (Hermann-Mauguin notation) list the essential symmetry elements along each of a crystal's principal directions — for example, **4/m32/m** is the full symbol for the highest-symmetry cubic class (a 4-fold axis with a perpendicular mirror, four 3-fold axes, and a 2-fold axis with a perpendicular mirror).

Exam shortcut

You rarely need to identify the full point group symbol for an entrance-exam MCQ — you almost always just need the **crystal system**, which depends only on the highest-order and most numerous rotation/rotoinversion axes present (see Section 5's diagnostic rules).

5. The Six Crystal Systems

Every crystal belongs to exactly one of six crystal systems, grouped by their **diagnostic (defining) symmetry** — the minimum symmetry every member of that system must possess.

Cubic (Isometric) System

Diagnostic symmetry: Always has **four** 3-fold (or 3) axes — the cube's body diagonals. May also have three 4-fold axes, six 2-fold axes, nine mirrors, and a center.

Axial relation: $a = b = c$, all angles = 90°.

Optical character: Isotropic (same optical properties in every direction).

Rule: "All crystals with four 3-fold axes of symmetry must belong to the cubic system." (Perkins, p. 302)

Hexagonal System

Diagnostic symmetry: A **single** 3-fold, 3, or 6-fold axis (never more than one). May also have 2-fold axes and mirrors.

Axial relation: $a = b \neq c$; angles 90° or 120°.

Optical character: Uniaxial — the single optic axis is parallel to the 3-, 3-, or 6-fold axis.

Rule: "Crystals with more than one 3-fold or 3 axis belong to the cubic system." (Perkins, p. 261)

Tetragonal System

Diagnostic symmetry: One (and only one) 4-fold or 4 axis.

Axial relation: $a = b \neq c$, all angles = 90° .

Rule: "Crystals that have more than one 4-fold axis must belong to the cubic system." (Perkins, p. 261)

Orthorhombic System

Diagnostic symmetry: Three mutually perpendicular 2-fold axes (222), or one 2-fold axis with two perpendicular mirrors (mm2).

Axial relation: $a \neq b \neq c$, all angles = 90° .

Monoclinic System

Diagnostic symmetry: A single 2-fold axis and/or a single mirror plane — lower symmetry than orthorhombic.

Axial relation: $a \neq b \neq c$; $\alpha = \gamma = 90^\circ$, $\beta \neq 90^\circ$.

Triclinic System

Diagnostic symmetry: At most an inversion center (I); no rotation axes higher than 1-fold, and no mirrors. The lowest-symmetry system.

Axial relation: $a \neq b \neq c$; none of α , β , γ equal 90° .

6. Crystallographic Axes & Unit Cell Parameters

Crystallographers describe a unit cell's shape with six parameters: three edge lengths **a**, **b**, **c** and three angles **α** (between b and c), **β** (between a and c), and **γ** (between a and b).

"The symmetry of a unit cell always affects the relationships between a, b, and c. So, in the cubic system, $a = b = c$, but in the tetragonal and hexagonal systems $a = b \neq c$."

— Perkins, *Mineralogy*, p. 307



Unit Cells, Points, Lines, and Planes

To describe the details of crystals and crystal structures, it is necessary to be more quantitative. In this chapter we describe points, lines, and planes in crystal structures using a coordinate system based on unit cell geometry. The most important results are that we can give exact descriptions of crystal structures, and we can label and describe crystal faces and crystal properties in unambiguous ways.

A crystal is composed of many identical building blocks called *unit cells*. To describe a unit cell, we must give its shape, size, and composition. Symmetry axes, crystal faces, and other linear and planar features of crystals depend on the nature of the unit cell and on crystal growth. To describe them, we need some way to describe the orientation of lines and planes in crystals. Furthermore, for complete characterization of a crystal, we must specify the locations of atoms in its structure. We therefore need a quantitative way to describe points (for example, atom locations), lines, and planes in crystals and unit cells. To do this, it is convenient to have a coordinate system. The system used by crystallographers is similar to a **Cartesian coordinate system**, with the exception that the angles between the three axes vary according to the crystal being described, as do the unit lengths along each axis.

UNIT CELL PARAMETERS AND CRYSTALLOGRAPHIC AXES

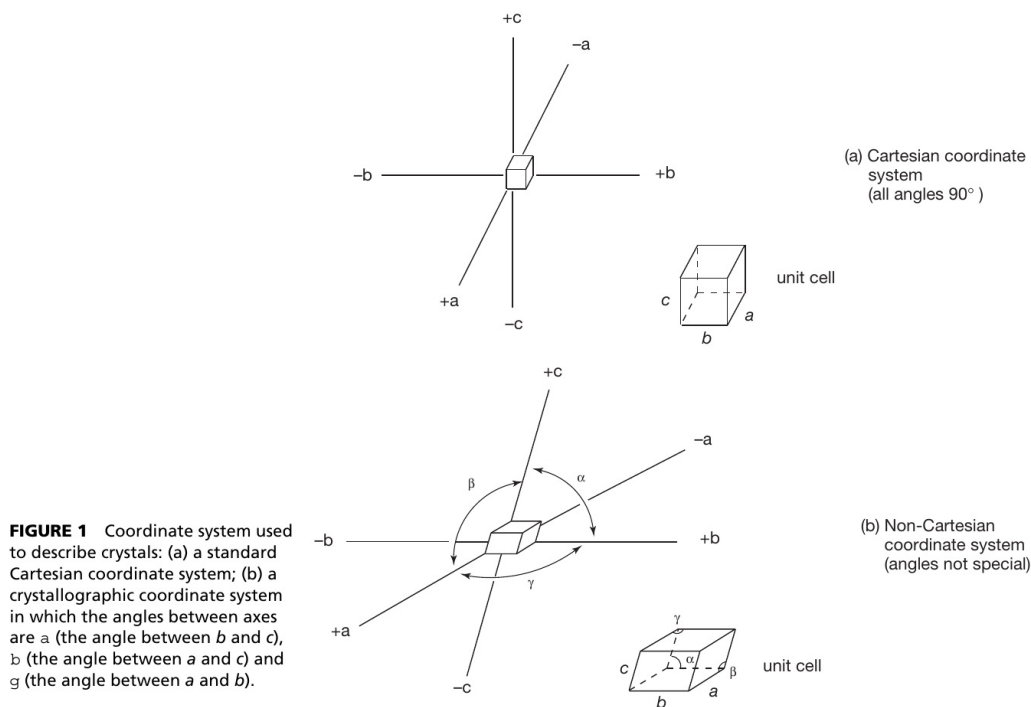
Unit cell parameters are a , b , c , α , β , and γ . a , b , and c are the lengths of unit cell edges; α , β , and γ are the angles between the edges, α being the angle between b and c , β the angle between a and c ,

and γ the angle between a and b . Mineralogists have traditionally used angstroms ($1 \text{ \AA} = 10^{-10} \text{ m}$) but some more recent literature uses nanometers ($1 \text{ nm} = 10 \text{ \AA}$) to measure a , b , and c . The angles α , β , and γ are today given as a decimal number of degrees (for example, 94.62°). Typical mineral unit cells have edges of 2 to $20 \text{ \AA}$; angles between edges vary greatly, although unit cells are often chosen so that angles are close to 90° .

Unit cell edges define the coordinate system used by crystallographers (Figure 1). Three axes, designated a , b , and c , run parallel to a , b , and c length edges of the unit cell. The angles between the axes are therefore α , β , and γ . In the cubic, tetragonal, and orthorhombic systems, α , β , and γ all equal 90° because the unit cell edges are orthogonal.

Instead of using angstroms to give distances, we may also use unit cell dimensions as a scale. For example, we might say that a certain plane intersects the axes at distances of $3a$, $2b$, and $2c$ from the origin. It is implicit that $3a$ refers to a distance equal to three unit cell edge lengths along the a axis, $2b$ a distance equal to two unit cell lengths along the b axis, and $2c$ a distance equal to two unit cell lengths along the c axis.

Figure 2, part 1 (Perkins, *Mineralogy*, pp. 307–308) — Unit cells of real minerals from the cubic, tetragonal, hexagonal, and orthorhombic systems, showing their axial lengths and angles.



The symmetry of a unit cell always affects the relationships between a , b , and c . So, in the cubic system, $a = b = c$, but in the tetragonal and hexagonal systems $a = b \neq c$. The relationships implied by crystal systems mean that, for systems other than triclinic, we need not give six values to describe unit cell shape. Table 1 gives examples of unit cell parameters for minerals from each of the crystal

systems; unnecessary information has been omitted. Figure 2 depicts the unit cells.

Consider the triclinic mineral albite, a feldspar with composition $\text{NaAlSi}_3\text{O}_8$. The last column in Table 1 gives albite's cell dimensions and Figure 2f shows a drawing of the unit cell with all edges and angles labeled. Because none of the angles are special and none of the cell edges are equal, we need six parameters to describe the unit cell shape. In

TABLE 1 Unit Cell Parameters (a , b , c , A , B , γ), Z (number of formulas per unit cell), and V (unit cell volume) for One Mineral from Each of the Crystal Systems*

Fluorite CaF_2	Rutile TiO_2	Beryl $\text{Be}_3\text{Al}_2\text{Si}_6\text{O}_{18}$	Enstatite $\text{Mg}_2\text{Si}_2\text{O}_6$	Sanidine KAlSi_3O_8	Albite $\text{NaAlSi}_3\text{O}_8$
cubic	tetragonal	hexagonal	orthorhombic	monoclinic	triclinic
$Z = 4$	$Z = 2$	$Z = 2$	$Z = 4$	$Z = 4$	$Z = 4$
$a = 5.46$	$a = 4.59$	$a = 9.23$	$a = 18.22$	$a = 8.56$	$a = 8.14$
$V = 162.77$	$c = 2.96$	$c = 9.19$	$b = 8.81$	$b = 13.03$	$b = 12.8$
	$V = 62.36$	$V = 2034.09$	$c = 5.21$	$c = 7.17$	$c = 7.16$
			$V = 836.30$	$\beta = 115.98$	$\alpha = 94.33$
				$V = 799.72$	$\beta = 116.57$
					$\gamma = 87.65$
					$V = 746.01$

* a , b , and c are in angstroms; α , β , and γ are in degrees; and V is in cubic angstroms.

Figure 2, part 2 (Perkins, *Mineralogy*, p. 308) — Monoclinic (sanidine) and triclinic (albite) unit cells — note progressively fewer axial and angular constraints as symmetry decreases.

7. Flow Chart: Identifying the Crystal System

This is the single most useful figure for solving MCQs like Q.16 — it converts a description of symmetry elements directly into a crystal system and point group.

BOX 6

Determining Crystal System and Point Group

The flow chart shown below can be used to determine a crystal's system and point group. The procedure is relatively straightforward for most well-formed crystals. However, for subhedral or anhedral crystals it is often impossible.

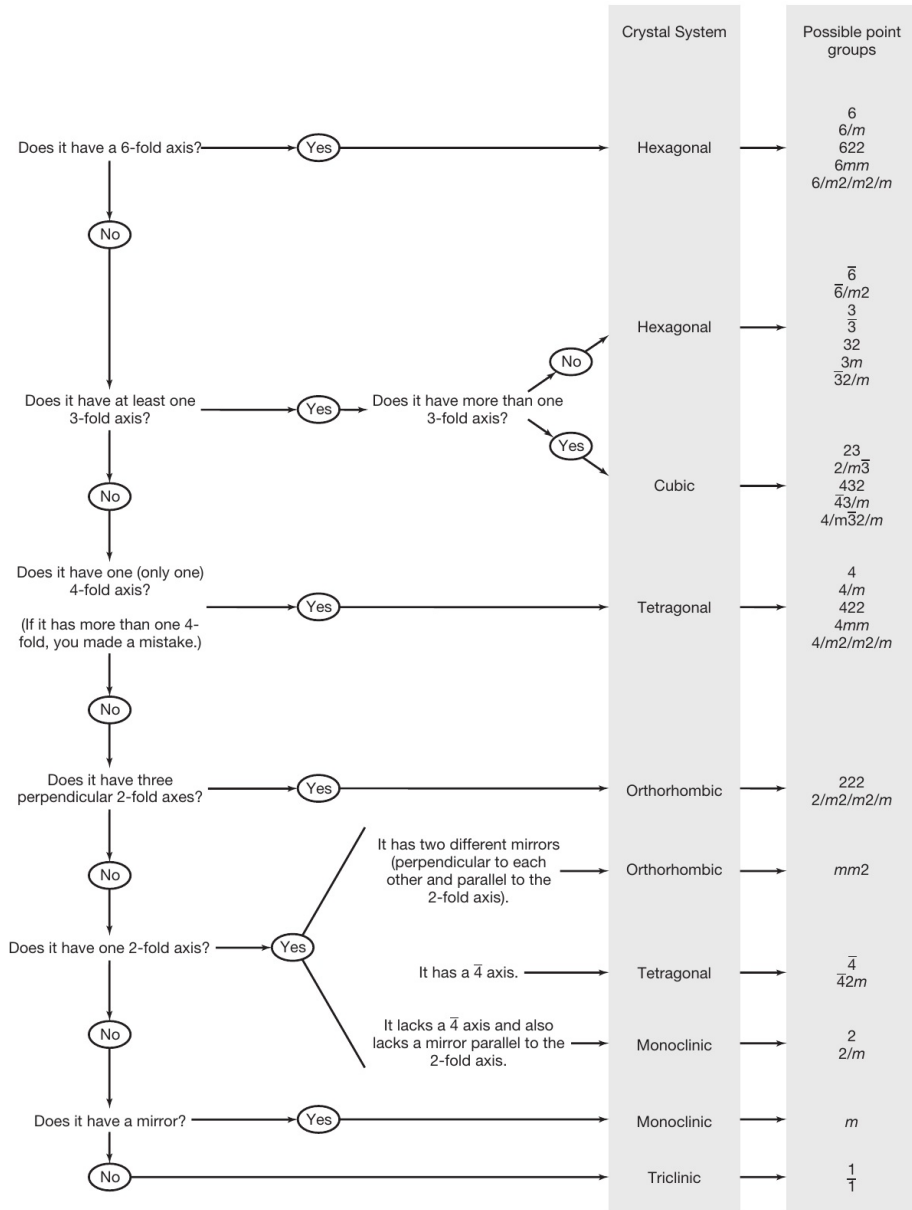


Figure 32 / Box 6 (Perkins, Mineralogy, p. 262) — Flow chart for determining crystal system and point group from observed symmetry elements.

8. Worked Strategy for Exam Questions Like Q.16

When a question describes a specific symmetry element (or combination) and asks for the crystal system, follow this sequence:

Step 1 — Count the highest-order axis and how many there are.

More than one 3-fold/3 axis → cubic, automatically. Exactly one 6-fold → hexagonal. Exactly one 4-fold → tetragonal (unless a second 4-fold appears, which forces cubic). Exactly one 3-fold/3 and no 4- or 6-fold → hexagonal.

Step 2 — If no 3-, 4-, or 6-fold axis is present, look at 2-fold axes and mirrors.

Three perpendicular 2-folds → orthorhombic. One 2-fold and/or one mirror → monoclinic. Only an inversion center or nothing → triclinic.

Step 3 — Check any stated angle against the special-angle table.

54°44' and 70°32' only arise from cube body/edge diagonals → cubic. 90°/60°/45°/30° correspond to the simple combinations

Step 4 — Eliminate distractor options using the "must belong to cubic" rules.

If an option like Hexagonal or Tetragonal is offered but the question specifies *more than one* high-order axis, it is automatically wrong — Perkins states plainly that multiple 3-fold or multiple 4-fold axes force the cubic system.

9. Quick-Reference Summary Table

System	Diagnostic Symmetry	Axial Relation	Angles
Cubic (Isometric)	Four 3-fold/3 axes	$a = b = c$	All 90°
Hexagonal	One 3-, 3-, or 6-fold axis	$a = b \neq c$	90° or 120°
Tetragonal	One 4-fold or 4 axis	$a = b \neq c$	All 90°
Orthorhombic	Three perpendicular 2-fold axes	$a \neq b \neq c$	All 90°
Monoclinic	One 2-fold axis and/or one mirror	$a \neq b \neq c$	$\alpha=\gamma=90^\circ, \beta\neq90^\circ$
Triclinic	At most an inversion center	$a \neq b \neq c$	None = 90°